Analytical Cume
February 12, 1994

One of the most sensitive analytical techniques available is fluorescence. A molecule absorbs light and emits light at an intensity proportional to the concentration of the molecule. The fluorescent light can also be quenched by the presence of other species. Often, the concentration of the quencher can be determined indirectly by monitoring the decrease in fluorescence it causes. Such is the case for the determination of fluoride in drinking water.

To four 5.00 mL aliquots of water sample were added 0.00, 1.00, 2.00, and 3.00 mL of standard NaF solution containing 10.00 ppb $F^-$. Exactly 5.00 mL of a solution containing an excess of Al-acid Alizarin Garnet R complex, a strongly fluorescing complex, were added to each of the solutions and diluted to 50.0 mL. The fluorescent intensity of the four solutions plus a blank are shown in the table below:

| $V_x$ | $V_s$ | I |
|---|---|---|
| 5.00 | 0.00 | 68.2 |
| 5.00 | 1.00 | 55.3 |
| 5.00 | 2.00 | 41.3 |
| 5.00 | 3.00 | 28.8 |

1. (10 points) Plot the data so that a straight line results using the graph paper provided.
2. (10 points) Determine the slope and intercept of the line
3. (5 points) Estimating the precision of the slope and intercept. You may use a calculator if you have one, (correlation coefficients are worthless) or you may estimate the uncertainty of each using the graph.

4. (35 points)
   Let $C_x$ = the concentration of $F^-$ in the water sample
   Let $C_s$ = the concentration of $F^-$ in the standard.
   Let $V_s$ = the volume of the standard added
   Let $V_x$ = the volume of the water sample used
   Let $V_t$ = the total volume of the solution analyzed.
   Let $I_x$ = the fluorescent intensity of the unknown solution analyzed
   Let $I_{x+s}$ = the fluorescent intensity of the unknown plus the an amount of standard added to the solution analyzed.

Derive the equation for the measured fluorescence in terms of the sample concentration and the standard concentration added and the volumes used.

5. (15 points) Show that when $I = 0$, $C_xC_x = C_sC_s$, and describe how the "x" intercept on the graph can be used to determine the concentration of $F^-$ in the sample, $C_x$

6. (25 points) Calculate the ppb of $F^-$ in the water sample, $C_x$ using the graphically determined slope and intercept. Give an uncertainty in the estimated value of $C_x$.

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1.(55 points) An analyst is presented with a sample containing approximately 10 different components which, based on solubility data, seem suited for separation by normal phase chromatography. It is desired to separate as many of these components as possible using an isocratic solvent system. Preliminary experiments show that all components are eluted with a k′ between 1 and 10 using a solvent composed of 55:45 diethyl ether: hexane but complete separation of all the components is not achieved.

Explicitly describe an experimental protocol using a four component solvent system, which will lead the analyst to the optimum solvent system for this separation. Show at least one of each type of calculation used to arrive at your final answer. Include in your protocol a brief discussion as to how the final optimized eluent will be chosen from your experimental data. The attached information will assist you in your write-up.

2.(45 points) Briefly discuss three of the following.

a. Affinity Chromatography
b. Capillary Electrophoresis
c. Ion chromatography
d. Ion exchange chromatography
e. Two dimensional TLC
f. Electrochemical detection for LC

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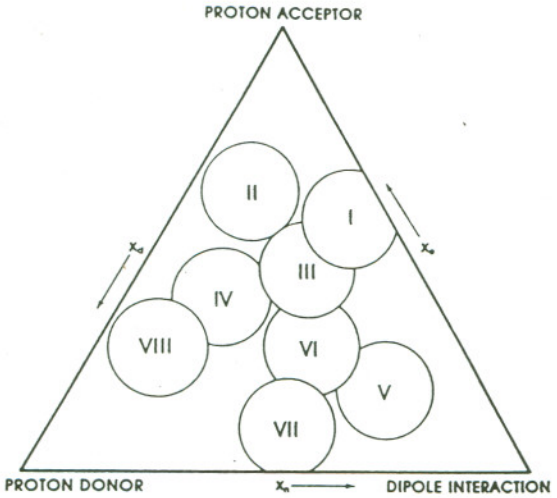
$$P_{AB} = \Phi_A P_A + \Phi_B P_B$$




*Grouping of solvents by selectivity.*

**Classification of solvent selectivity[4]**

| Group | Solvents |
|---|---|
| I | Aliphatic ethers, tetramethylguanidine, hexamethyl phosphoric acid amide (trialkyl amines) |
| II | Aliphatic alcohols |
| III | Pyridine derivatives, tetrahydrofuran, amides (except formamide) glycol ethers, sulfoxides |
| IV | Glycols, benzyl alcohol, acetic acid, formamide |
| V | Methylene chloride, ethylene chloride |
| VI | a) Tricresyl phosphate, aliphatic ketones and esters, polyethers, dioxane |
| | b) Sulfones, nitriles, propylene carbonate |
| VII | Aromatic hydrocarbons, halosubstituted aromatic hydrocarbons, nitro compounds, aromatic ethers |
| VIII | Fluoroalkanols, m-cresol, water (chloroform) |

**Solvent characteristics[4,6]**

| Solvent | Snyder (Figure 1) selectivity group | Strength weighing factor ($s_j$) (RP) | (NP) |
|---|---|---|---|
| Methanol | II | 2.6 | 5.1 |
| Acetonitrile | VI | 3.2 | 5.8 |
| Tetrahydrofuran | III | 4.5 | 4.0 |
| Water | — | 0 | 10.2 |
| Chloroform | VIII | | 4.1 |
| Methylene chloride | V | | 3.1 |
| Methyl-t-butyl ether | I | | ~2.5 |
| Ethyl ether | I | | 2.8 |
| Hexane | — | | 0 |

$$\text{Resolution } R_s \cong \frac{1}{4}(N)^{1/2} \cdot \frac{k'}{1+k'} \cdot \frac{(\alpha-1)}{\alpha}$$